

Decoupling Extensions from Chrome

The Journey to a Modular Extensions Layer

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About us

- We are part of Igalia's Chromium team
- Focus on open source browsers for the last two decades



This talk presents the ongoing effort to decouple Chromium Extensions from the Chrome layer, enabling a more modular and embeddable architecture. We'll cover the motivation, migration strategy, lessons learned, and current progress, along with a live demo.



Context and motivation

- The problem to solve: extensions support in non-desktop/embedded scenarios
 - Modular and reusable extension architecture
 - Better maintainability and portability
 - Reduce monolithic //chrome specific coupling
- Targeting platforms as embedded Linux SoC (Raspberri Pi, eg), iOS, or Android mobile devices
- Also smaller components based on minimal Chromium builds

We identify scenarios of companies having this kind of specific needs, for which Chromium architecture was not entirely suitable.

Ongoing timeline

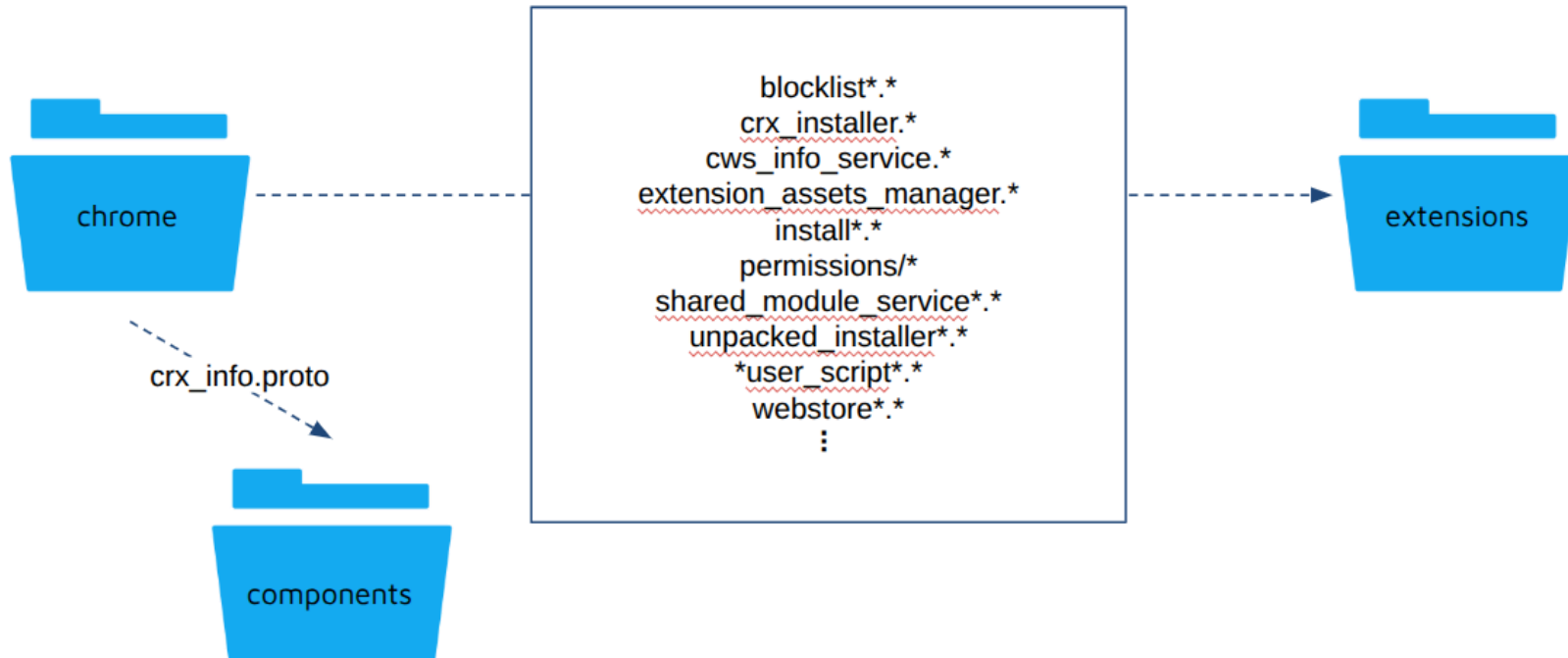
- Enablement for Extensions on Android in downstream project (2023.04)
- Present Extensions on Android BlinkOn 20 (2025.04)
- Supported partially Extensions in Wolvic Project (after BlinkOn 20)
- Started migrating to `//extensions` (2025.09)
- Shared updated status report in BlinkOn 21 (2026.04)

Status of CLs

- Merged : 48, Reviewing : 2

☐	Subject	Owner	Reviewers	Repo	Branch	Updated	Size	Status	CR	V	CQ
☐	☆ Move SupervisedUserExtensionsMetricsRecorder to //extensions	Miyoung Shin	Devlin (C... , Thomas	chromium/src	main	8:33 AM	M	Merged	✓		+2
☐	☆ Reduce Profile dependency in webstore_private_api.cc	Miyoung Shin	Devlin (C...	chromium/src	main	May 21	M	Merged	✓		+2
☐	☆ Move extension_install_status.{cc,h} to //extensions	Miyoung Shin	Devlin (C... , Kyle	chromium/src	main	May 19	L	Merged	✓		+2
☐	☆ Move webstore_install_result.{cc/h} to //extensions.	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	May 11	S	Merged	✓		+2
☐	☆ Move webstore_override.{cc/h} to //extensions.	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Apr 24	S	Merged	✓		+2
☐	☆ Move Scripting API to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 24	S	Merged	✓		+2
☐	☆ Move kCloudExtensionRequest{Ids Enabled} to //components/enterprise	Miyoung Shin	Devlin (C... , Nicolas	chromium/src	main	Apr 20	M	Merged	✓		+2
☐	☆ Move key names used by prefs::kCloudExtensionRequestIds to //extensions	Miyoung Shin	Devlin (C... , Nicolas	chromium/src	main	Apr 17	S	Merged	✓		+2
☐	☆ Move ManifestV2ExperimentManager to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 16	M	Merged	✓		+2
☐	☆ Move WebstoreInstallHelper to //extensions	Miyoung Shin	Devlin (C... , Mike West, +1	chromium/src	main	Apr 16	M	Merged	✓		+2
☐	☆ Move messages for Web Store in generated_resources.grd to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 16	S	Merged	✓		+2
☐	☆ Replace BitmapFetcher with ImageFetcher in WebstoreInstallHelper	Miyoung Shin	Devlin (C... , Sergii	chromium/src	main	Apr 16	M	Merged	✓		+2
☐	☆ Move managed_installation_mode.h to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Apr 15	S	Merged	✓		+2
☐	☆ Move MV2DeprecationImpactChecker to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 15	M	Merged	✓		+2
☐	☆ Move ExtensionAllowlist to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 14	M	Merged	✓		+2
☐	☆ Move mv2_experiment_stage.h to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 14	S	Merged	✓		+2
☐	☆ Remove GetPrefServiceForContext from ExtensionsBrowserClient	Miyoung Shin	Devlin (C...	chromium/src	main	Apr 12	M	Merged	✓		+2
☐	☆ Move CWSInfoService to //extensions.	Miyoung Shin	Devlin (C... , Hidehiko, +1	chromium/src	main	Apr 01	M	Merged	✓		+2
☐	☆ Move webstore_data_fetcher.{cc/h} to //extensions.	Miyoung Shin	Devlin (C... , Mike West, +2	chromium/src	main	Mar 30	S	Merged	✓		+2
☐	☆ Move webstore_data_fetcher_delegate.{cc/h} to //extensions.	Miyoung Shin	Devlin (C... , Sergii	chromium/src	main	Mar 27	S	Merged	✓		+2
☐	☆ Use extensions::util::IsExtensionDownload directly	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Mar 25	M	Merged	✓		+2
☐	☆ Move WebstoreInstaller to //extensions.	Miyoung Shin	Avi, Devlin (C... , +1	chromium/src	main	Mar 24	L	Merged	✓		+2
☐	☆ Move cws_item_service.proto to //extensions.	Miyoung Shin	Avi, Devlin (C... , +1	chromium/src	main	Mar 23	M	Merged	✓		+2
☐	☆ Move kAppsGalleryDownloadURL to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Mar 21	S	Merged	✓		+2
☐	☆ Move kCWSInfo* preferences to //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Mar 21	S	Merged	✓		+2
☐	☆ Move SharedModuleService to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Mar 18	M	Merged	✓		+2
☐	☆ Move CrxInstaller to //extensions.	Miyoung Shin	Devlin (C... , Hidehiko	chromium/src	main	Feb 24	L	Merged	✓		+2
☐	☆ Move kAppsGalleryURL to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Feb 19	M	Merged	✓		+2
☐	☆ Move messages in generated_resources.grd to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Feb 10	M	Merged	✓		+2
☐	☆ Move GetFixupExtensionNameForUIDisplay() to //extensions	Miyoung Shin	Avi, Devlin (C...	chromium/src	main	Feb 10	M	Merged	✓		+2
☐	☆ Introduce ExtensionInstallPromptClient in //extensions	Miyoung Shin	Devlin (C...	chromium/src	main	Feb 03	M	Merged	✓		+2
☐	☆ Replace Profile with BrowserContext in CrxInstaller	Miyoung Shin	Devlin (C...	chromium/src	main	Jan 26	M	Merged	✓		+2
☐	☆ Use ScopedExtensionKeepAlive in CrxInstaller	Miyoung Shin	Devlin (C...	chromium/src	main	Jan 22	M	Merged	✓		+2
☐	☆ Relocate ScopedExtensionUpdaterKeepAlive	Miyoung Shin	Devlin (C...	chromium/src	main	Jan 22	M	Merged	✓		+2
☐	☆ Move InstallStageTracker to //extensions.	Miyoung Shin	Avi, Devlin (C... , +1	chromium/src	main	Jan 18	L	Merged	✓		+2
☐	☆ Replace Profile with BrowserContext in InstallApproval	Miyoung Shin	Devlin (C...	chromium/src	main	Jan 16	S	Merged	✓		+2

Migrated Files



What Changed?

- Migrated files related to installation
- Support all extension formats on `//extension` layer
 - CRX – the standard packaged format used by the Chrome Web Store
 - Unpacked – primarily used during development and debugging
 - Zip – commonly used for internal distribution or packaged deployment

Demo

- We are showing now a demo video of the support on the app_shell target
- Support for installing and running some demo extensions

What Next?

- Complete the migration of Web Store-related code
 - Allow embedders to install extensions directly from the Chrome Web Store
 - Continue migrating APIs from the `//chrome` layer to `//extensions`
 - Completed: Script API
 - In Progress: WebStore Private API
 - Future candidates: action, cookies, dom, tabs, windows and other commonly used extension APIs
- => Allow embedders to be able to run commercial extensions from Chrome Web Store

Challenges

- API Migration Complexity
 - Script API → 1 day
 - WebStore Private API → several months
 - UI-heavy APIs (e.g. action) remain difficult because of deep `//chrome` dependencies
- Testing Architecture
 - all test in `//extensions` have moved into `//chrome`
 - This makes it difficult to validate functionality that is intended to live in `//extensions`
 - This works against the goal of making extensions more embedder-friendly
- Demo & Validation Environment
 - Extension Shell was removed from ToT after all tests were migrated to `//chrome`
 - It was a useful lightweight environment built on Content Shell
 - We now need alternative ways to demonstrate and validate extension support

More information

- Blog posts
 - [Our journey to support extensions for embedders](#)
 - [Extension migration progress update](#)
- Videos
 - [BlinkOn 21 presentation](#)

Thanks!!